

Generate and distribute electrical power to meet uncertain demand.

Systems Modeling for Climate Response

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In this project you will optimize the distribution of electrical power within a grid and assess the risk of failing to satisfy electrical demand due to uncertainties in capacity and demand. In the optimization stage you will compare results computed using linear programming and quadratic programming. In the risk analysis stage you will compare results that involve correlated uncertainties with results that neglect correlations.

1 Case Studies

1.1 Case Study: The Valentine's Day Texas Power Grid Emergency

In February of 2021, The American South, including all of Texas, was hit with anomalously cold winter weather. This climate crisis was made further by widespread blackouts to the Texas power grid, resulting in the Texas Power Grid Emergency. The majority of Texas is serviced by a single wide area interconnection (the technical term for a power grid). In general, a wide area interconnection is the solution to the number one problem surrounding supply and demand of electricity: volatility. Because electricity cannot be stored, must be produced, transmitted, and consumed in the same instant; this is why a large, robust power grid with exposure to a lot of different geographies, power generation types, etc. in aggregate creates resilience. It is very important that the frequency of the power grid remains within an acceptable range or the various elements connected to the grid risk being damaged. The frequency is determined by the supply-demand balance. Demand is changing all the time, so the party overseeing the grid must change the power generation at the same time in order to keep the supply and demand in sync. The Texas power grid is overseen by ERCOT, a non-profit corporation governed by a board of stakeholders from all parts of the grid. ERCOT doesn't own or operate any part of the grid but rather it tells operators when to start and stop running in order to keep supply and demand in sync. It also manages scheduled outages and oversees the market by setting prices and managing transactions between buyers and sellers.

The week before the Texas Power Grid Emergency had already been a wintry one, and weather forecasts predicted extreme cold the next week. ERCOT anticipated the massive demand and canceled planned maintenance outages in anticipation. Despite this planning, the extreme weather prevailed. The storm that eventually hit on Valentine's day was so devastating for three reasons: 1) the extremely cold temperatures 2) its wide-ranging effects—it impacted the entire state and beyond 3) the duration of the frigid temperatures was so long. Texas reached its record peak winter electrical demand of 70,000 MW. The Texas Power Grid has a diverse portfolio of power generators by type, but this diversification did

not save the grid. Wind and natural gas made up the majority of the lost capacity. Natural gas alone lost 15,000 mW of capacity in the span of eight hours. The pipelines themselves are vulnerable to temperature, and when the pipelines froze, the roads to fix them were inaccessible. Even if natural gas power plants remained online, many were financially unable to operate because the wholesale price of natural gas spiked 100x during the period of the storm. Wind power shut off due to icing concerns, solar panels were covered in snow, and even one of Texas' nuclear power plants was shut off for cold weather concerns surrounding a frozen water pipeline.

Compared to other power grids, Texas' is relatively independent. This put it in less of a position to rely on other systems to help balance supply and demand. And while the Texas power grid does have a few connections to other power grids, these other grids (all geographically adjacent) were similarly affected by the wide breadth of the storm, leaving them without extra electricity to spare. In order to prevent permanent damage to the grid's components, ERCOT had one choice to keep supply and demand in sync: "shed load" or turn off customers' access to power. This left millions without power.

The energy market is an important part of the story. Unlike other markets that pay generators to secure capacity for the future, the wholesale electricity market in Texas is energy-only, meaning if you put power on the grid, you don't get compensated for capacity when it isn't needed. This means the generators do not have the same incentive to invest in the resiliency of the grid. [1]

1.2 Case Study: The 2003 Northeast Blackouts

Beginning on August 14, 2003, approximately 50M people from Ottawa, Canada to Cleveland, OH, to New York, NY were without power. All parts of life were disrupted. The most immediate concerns of emergency responders were the people stuck in elevators and underground subways. Attention quickly shifted to the water supply as the flow of clean water is dependent on electric pumps. Two hundred and sixty five power plants were affected. It took anywhere from two to 29 hours to restore power to the major cities; in some places, it took two weeks for the power to come back. [2] Overall, at least 11 deaths and \$6.4B in economic losses are attributed to the blackout.

After the crisis was solved, engineers went to work to try to figure out what happened. Terrorism was quickly ruled out. It took engineers months to unravel the cause of the blackout, but it was ultimately traced back to a downed power line in Ohio. A power line in suburban Cleveland sagged due to excess current and intercepted a tree that should have been better trimmed. The initial problem was mostly the fault of FirstEnergy in Akron. [3] From there, a cascading effect of human and equipment failures meant that the problem wasn't caught and lines continued overloading farther and farther away. An employee who worked for MISO - the overseeing agency - went to lunch and forgot to turn back on a tool that monitored grid problems. Eventually, the grid in Ohio got less and less stable and voltage became too much and plants farther and farther away began shutting down. The outage ended up taking down the entire Eastern Interconnection, one of three in the United States (Eastern, Western, and the aforementioned Texas). The highest impact outcome of the blackout was the making of industry standards mandatory and enforceable. [4]

2 Problem Statement

In this project, you will design the most economical distribution of electrical power within a power grid and will assess the robustness of the power grid to uncertain extreme electrical power demands, while considering limitations on the capacity for power generation and uncertainty in the power transmission capacities of the grid.

The topology of the power grid is shown in Figure 1.

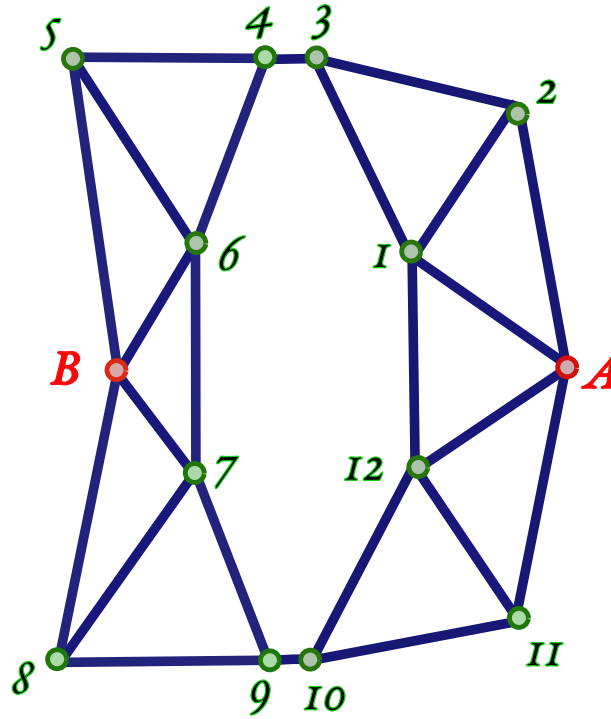


Figure 1. Power grid topology. Nodes A and B are power generating nodes and consume no power. Nodes 1 through 12 are power demand nodes (cities, factories, etc) and at any point in time demand a certain amount of power. The generating nodes and the demand nodes are interconnected as shown.

This system can be represented by twenty-four individual connections, eight connecting station to substation and sixteen between substations. The design variables are this set of 24 power flows, $P_{i,j}$. For consistency, $P_{i,j}$ represents the power flow *from* node i *to* node j where $i < j$. Order the 24 power flow design variables as follows ...

$$\mathbf{x} = \begin{bmatrix} P_{A,1} & P_{A,2} & P_{A,11} & P_{A,12} & P_{B,5} & P_{B,6} & P_{B,7} & P_{B,8} & \dots \\ P_{1,2} & P_{1,3} & P_{1,12} & P_{2,3} & P_{3,4} & P_{4,5} & P_{4,6} & P_{5,6} & P_{6,7} & P_{7,8} & P_{7,9} & P_{8,9} & P_{9,10} & \dots \\ P_{10,11} & P_{10,12} & P_{11,12} \end{bmatrix}^T$$

Any feasible distribution of electrical power adheres to the constraints that:

1. The total power generated at a generating station can not exceed the power generating capacity of that station.
2. The total power delivered to a demand node must not be less than the power demanded at that node.
3. The power transmitted along a line must be within the power carrying capacity of the line.
4. Generating nodes must output power, they may not receive power.

The analysis of a power grid comes down to balancing the input power and output power of the grid. For example, at node 6,

$$P_{B,6} + P_{4,6} + P_{5,6} - P_{6,7} \geq P_{6,D}$$

(assuming that $P_{i,j} > 0$ represents power flow *from* node i *to* node j where $i < j$, and that power demanded by node 6 ($P_{6,D}$) leaves node 6.)

Denote the power generating capacities of stations A and B as G_A and G_B .

Denote the power transmission capacities of each line as $T_{i,j}$.

For lines connected to power generating stations, $0 \leq P_{i,j} \leq +T_{i,j}$.

For all other lines, $-T_{i,j} \leq P_{i,j} \leq +T_{i,j}$

The cost of generating electricity f , is to be minimized.

$$f(\mathbf{x}) = c_A P_{A,1} + c_A P_{A,2} + c_A P_{A,11} + c_A P_{A,12} + c_B P_{B,5} + c_B P_{B,6} + c_B P_{B,7} + c_B P_{B,8}.$$

3 Tasks

In Tasks 1, 2, 3, and 4 analyze the power grid as a linear program (LP); solve for the optimal power grid distribution in Tasks 5, 6, and 7; and assess the risk of a blackout in Tasks 8 and 9.

1. Write the 14 inequalities corresponding to constraints (1.) and (2.) above for generating stations A , B , and substations 1 through 12.
2. Express these 14 inequalities in matrix form $\mathbf{Ax} \leq \mathbf{b}$ in which the 14-by-24 matrix $\mathbf{A}$ will contain values of 0, 1, and -1.
3. Express the objective function in vector form $f = \mathbf{c}^\top \mathbf{x}$
4. Write lower bounds and upper bounds for the power transmission
5. Here is a general purpose `.m`-function for solving *any* LP by using `SQPopt`.
Just type the following six lines into a `.m`-file called `... LP_analysis.m`.

```

1 function [f,g] = LP_analysis(x,constants)
2 % [f,g] = LP_analysis(x,constants)
3 % analyze a trial solution x to a linear programming problem,
4 % minimize f = c'*x such that g = A*x-b <= 0
5
6 A = constants{1}; % constraint coefficient matrix ( dimension m by n )
7 b = constants{2}; % constraint vector ( dimension m by 1 )
8 c = constants{3}; % cost coefficient vector ( dimension n by 1 )
9
10 f = c'*x; % the cost function
11
12 g = A*x-b; % the constraint inequalities , compared to zero

```

Why must there be curly brackets in this code?

6. Complete the .m-file shown on page 6. In matrix **A**, use values of 0, 1, and -1. In vectors **b**, **c**, use variables **G**, **D**, **cA** and **cB** (not numerical values). In vectors **x_lb** and **x_ub** use 0 and variable **T**. And in scalars **netGeneration**, and **netDemand** use variables **D** and **x_opt**.
7. This code provides three numerical methods for solving the LP:
 - **linprog** using the dual-simplex algorithm
 - **linprog** using the interior-point algorithm
 - **SQPopt**

Solve the problem with each of the three methods. Using **linprog** in Matlab requires the Optimization Toolbox. Typing **help linprog** at the **>>** prompt will indicate whether or not you have the Optimization Toolbox.

- (a) For each of the three solutions
 - i. Note the value of **f_opt** and **x_opt**
 - ii. Confirm that $\mathbf{Ax} - \mathbf{b} \leq \mathbf{0}$
 - iii. Confirm that $-T_{i,j} \leq P_{i,j} \leq T_{i,j}$.
 - iv. Calculate the power shortfall ... **netDemand** - **netGeneration**
- (b) Which optimization method results in the lowest cost?
- (c) Which optimization method results in the design that *you* think is best?
Why do you think it is best?

What you find depends on how you look for it.

```

1
2 G = [ 60 ; 225 ];           % power generating capacities of plants A and B
3 cA = 20;                   % smaller plant, less capacity, less efficient
4 cB = 12;                   % larger plant, more capacity, more efficient
5
6 A = [ ...                  % matrix inequality coefficient matrix (m x n)
7       ... ];
8
9 [m,n] = size(A);           % m : number of constraints , n : number of variables
10
11 %           1   2   3   4   5   6   7   8   9   10  11  12
12 D = [ 30 ; 20 ; 5 ; 7 ; 35 ; 25 ; 25 ; 20 ; 5 ; 6 ; 25 ; 30 ]; % 12 power demands
13 b = [ ...                  % matrix inequality vector b (m x 1)
14       ... ];
15 c = [ ... ];               % cost function coefficients vector c (n x 1)
16
17 T = 48*ones(n,1);          % every line has the same capacity (n x 1)
18 x_lb = [ ... ];            % lower bound of the solution, x (n x 1)
19 x_ub = [ ... ];            % upper bound of the solution, x (n x 1)
20
21 sumA = sum(A)               % check columns of A sum to zero
22
23 % solve the problem as an LP ... matlab's linprog
24 % {
25 disp('linprog -----')
26 % use one of these two algorithm options
27 linprog_options = optimoptions('linprog','Algorithm','dual-simplex');
28 %linprog_options = optimoptions('linprog','Algorithm','interior-point');
29 [ x_opt, f_opt, ~,~, y_opt ] = ...
30     linprog( c, A, b, [], [], x_lb, x_ub, linprog_options);
31 x_opt
32 f_opt
33 g_opt = A*x_opt - b
34 y_opt = y_opt.ineqlin
35 netGeneration = ...
36 netDemand = ...
37 % }
38
39 if (input(' OK to continue? [y]/n : ','s')== 'n') return; end
40
41 % solve the problem as an SQP ... hpg's SQPopt
42 % {
43 disp('SQPopt -----')
44 constants = { A , b , c }; % the set of constants in this problem
45
46 x_init = ... % initial guess for x ... start near zero for everything (n x 1)
47
48 %           msglev  tolX  tolF  tolG  MaxEvals
49 options = [      0    0.01  0.01  0.01    1e4      ];
50
51 [x_opt,f_opt,g_opt,cvg_hst,y_opt] = ...
52     SQPopt('LP_analysis',x_init,x_lb,x_ub,options,constants);
53
54 y_opt = y_opt(1:length(b))
55 netGeneration = ...
56 netDemand = ...
57 % }

```

8. Now consider the reliability of the power grid to uncertain demand extremes and uncertain line capacity.

Power demands $P_{i,D}$ are uncertain. Spiking demands are a known challenge to the grid. In this project the demands are modeled as a vector of 12 lognormally distributed random variables with median values as listed in the vector **D** (in the code above) and with a coefficient of variation of 0.1 for all nodes. To start with, assume that the variation of any demand ($P_{i,D}$) has no bearing on the variation of any other demand (and vice-versa). In other words, all pairs of demands $P_{i,D}$ and $P_{j,D}$ are *uncorrelated*. Download [corr_logn_rnd.m](#) and add code like this to your .m-script from page 6.

```

1 % statistical modeling of each demand
2 medD = D; % median demand
3 nd = length(D); % number of demand nodes
4 covD = 0.10*ones(nd,1); % coefficient of variation of demand
5 RD = eye(nd); % correlation matrix for task 8 : no correl.
6 Drand = corr_logn_rnd( medD, covD, RD, N ); % nd-by-N matrix of random D

```

Similarly line capacities T are uncertain. Loss of transmission line capacity is another known challenge. In this project the line capacities are modeled as a vector of n beta-distributed random variables with capacities between 30 MW and 65 MW, and with lower and upper exponents $q = 4$ and $p = 2$. To start with, assume that the variation in any line capacity $T_{i,j}$ has no bearing on the variation of any other line capacity (and vice-versa). In other words, all pairs $T_{i,j}$ and $T_{k,l}$ are *uncorrelated*. Download [corr_beta_rnd.m](#) and add code like this to your .m-script.

```

1 % statistical modeling of transmission capacity in each line
2 aT = 30.0*ones(n,1); % lower transmission capacity, MW
3 bT = 65.0*ones(n,1); % upper transmission capacity, MW
4 qT = 4.0*ones(n,1); % 'lower' exponent of Beta distribution
5 pT = 2.0*ones(n,1); % 'upper' exponent of Beta distribution
6 RT = eye(n); % correlation matrix for task 8 : no correl.
7 Trand = corr_beta_rnd( aT,bT, qT,pT, RT, N ); % correlated beta sample

```

- (a) Refer to sections 4, 5.2, p. 10, p. 13, sections 12.2, 12.3 and 17 of [Probability Distributions](#) for background. Use the first block of 6 lines of code above to generate a sample of $N=100$ values of **nd** uncorrelated random demands, distributed as log-normal random variables with median values as listed in the vector **D** (of the code above) and coefficient of variation of 0.1 for all demand values. Use the second block of 7 lines of code above to generate a sample of $N=100$ values of **n** uncorrelated random line capacities, distributed as beta random variables between 30 and 65 MW.
- (b) For each of the $N=100$ Monte Carlo simulations
 - i. Update the vectors **b**, **x_lb** and **x_ub** with a random vector from the sample of **Drand**, **Trand** and **consts** generated above.
 - ii. Use **SQPopt** to attempt to find a feasible solution to the power generation and distribution problem.
 - iii. Keep a record of the net power generated and the net power demanded for each sample of the N values, as shown at the end of the code on page 6.

- (c) Use `plotCDFci.m` to plot the cumulative distribution of the sample one of the twelve demands. Does the median value of the sample match the specified median value for the lognormal distribution?
 - (d) Likewise for one of the `n` line capacities. Compare the max and min values of the sample to the max and min values specified for the beta distributions.
 - (e) Make a scatter plot of the sample of the demand in one node vs the demand in another node. Do these demands appear uncorrelated?
 - (f) Likewise for a pair of line capacities.
 - (g) Use `plotCDFci.m` to plot the cumulative distribution of net demand, net generation, and shortfall ... (shortfall = net demand - net generated). Which distribution looks most like a normal distribution? Which distribution looks least like a normal distribution?
 - (h) For what fraction of this sample is the shortfall greater than 0.1 MW?
 - (i) Is the unmet demand due to an overloaded transmission line or insufficient generation capacity (or both)?
9. Repeat step 8 but with correlated demand and correlated line capacity. Positively correlated demand models the likely situation that when one node has higher (or lower) demand other nodes are also likely to have higher (or lower) demand ... like in a summer heat wave. Likewise for the line capacity. The probabilities are *conditionally dependent*. This is easily implemented in the code above. For correlated demands, change the power demand correlation matrix `RD` in task 8 to ...

```
1 RD = 0.8*ones(nd) + 0.2*eye(nd); % correlated variability between node demands
```

For correlated line capacities, change the line capacity correlation matrix `RT` in task 8 to ...

```
1 RT = 0.7*ones(n) + 0.3*eye(n); % correlation matrix for task 9 : pos. correl.
```

This means that the correlation in the demand between any pair of stations is 80 percent and that the correlations in demand between any pair of transmission lines is 70 percent.

Do correlations among uncertain system attributes affect the reliability of the system?

To hand in:

1. (12 points) Your hand written work for tasks 1 through 4.
2. (10 points) Your answer to the question in task 5.
3. (10 points) Your code for task 6.
4. (11 points) Your values of `f_opt` and `x_opt` for each of the three numerical methods in task 7. Statements for parts ii., iii., and iv., in task 7. Written answers to 7.(a) and 7.(b).
5. (10 points) Your code for tasks 8. and 9. These codes will differ by one line as indicated in task 9.
6. (11 points) Your plots for 8.(c) - 8.(g) with axis labels. Your answers for 8.(h) and 8.(i).
7. (11 points) Your plots for 9.(c) - 9.(g) with axis labels. Your answers for 9.(h) - 9.(i).
8. (10 points) Your answer to the last question in task 9.
9. (15 points) Neatness.

4 References

1. <https://www.youtube.com/watch?v=08mwXICY4JM>
2. https://www.youtube.com/watch?v=FMCPuE_7oE8
3. <https://www.youtube.com/watch?v=REcl2ly34hg>
4. <https://www.scientificamerican.com/article/2003-blackout-five-years-later/>
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6. Ardelean M and Minnebo P. HVDC Submarine Power Cables in the World: State-of-the-Art Knowledge. EUR 27527. Luxembourg (Luxembourg): Publications Office of the European Union; 2515. JRC97720